



Machine learning-driven prediction of magnetic parameters in magnetic nanoparticles

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ABSTRACT

Magnetic nanoparticles (MNPs) have attracted considerable attention for applications in biomedicine, data storage, and catalysis. Their magnetic behavior is governed by three key hysteresis parameters: coercivity (H_c), remanent magnetization (M_r), and saturation magnetization (M_s). Traditional theoretical models, such as the Stoner-Wohlfarth model, cannot adequately capture nanoscale phenomena including surface spin disorder, thermal fluctuations, and interparticle interactions. This study develops a machine learning framework to simultaneously predict these three parameters. A curated dataset of 1,758 high-quality samples was compiled from representative materials (Fe_3O_4 , Fe_2O_3 , and CoFe_2O_4). Key descriptors include geometry, chemical composition, crystal structure, and synthesis conditions. We evaluated multiple machine learning models. Among them, Gradient Boosting, XGBoost, and ExtraTrees achieved the highest coefficient of determination (R^2) values of 0.78, 0.85, and 0.93 for H_c , M_r , and M_s , respectively. SHapley Additive exPlanations (SHAP) analysis revealed the dominant factors controlling each parameter. Exchange bias field primarily influences H_c . M_r is directly influenced by M_s and H_c , with chemical composition (Fe, Co, Ni, O ratios) also playing an important role. M_s is primarily influenced by core magnetic moment and particle geometry. These findings uncover strong intrinsic coupling among the three parameters, particularly the direct influence of M_s and H_c on M_r . This work establishes a robust predictive framework for performance optimization and rational design of magnetic nanoparticles.

1. Introduction

1.1. Background and Significance

Magnetic nanoparticles (MNPs) possess unique size effects, surface effects, and quantum confinement phenomena, making them one of the most promising functional materials of the twenty-first century [1–3]. They have widespread applications in biomedicine, data storage, and environmental remediation. In magnetic hyperthermia, particles generate heat under alternating magnetic fields. Therapeutic efficacy depends on the balance between coercivity (H_c) and saturation magnetization (M_s). In magnetic resonance imaging (MRI), contrast enhancement efficiency is primarily determined by M_s [2,4]. In magnetic separation and drug delivery, simultaneous optimization of H_c , M_s , and remanent magnetization (M_r) is essential [4,5]. Achieving precise control over these magnetic properties requires a deep understanding of three key hysteresis parameters.

These three key hysteresis parameters are fundamentally important for magnetic nanoparticle applications. H_c represents the magnetic field strength required to reduce magnetization to zero under a reverse field, indicating the material's resistance to demagnetization. Materials with high H_c are suitable for permanent magnets, while low- H_c materials are preferred for soft magnetic applications. M_r is the residual magnetization retained after removing the external field, reflecting magnetic memory capability. M_s represents the maximum magnetization when all magnetic moments are fully aligned, directly influencing performance in MRI, magnetic hyperthermia, and separation. Importantly, M_r and H_c jointly determine hysteresis loss characteristics critical for magnetic fluid hyperthermia, while M_s sets the upper limit for M_r . Together, these parameters form a comprehensive framework for describing magnetic behavior.

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1.2. Limitations of traditional theoretical approaches

Traditional prediction methods face significant limitations for complex nanoscale systems. The Stoner–Wohlfarth model assumes coherent rotation. It neglects surface spin disorder and interparticle interactions [6]. The Néel–Brown model assumes a single energy barrier, making it inadequate for describing complex structures such as core–shell and multilayered systems [7]. First-principles calculations provide atomistic accuracy. However, they require thousands of central processing unit (CPU) hours per system. Their reliance on periodic boundary conditions limits capturing surface reconstruction and finite-temperature effects. They also introduce systematic errors in strongly correlated systems [8]. Micromagnetic simulations assume spatially homogeneous material parameters. This assumption is inconsistent with compositional fluctuations and grain boundary effects in real nanoparticles. Their accuracy depends critically on the input parameters, creating a circular dependency [9].

1.3. Machine learning approaches in magnetic materials

Machine learning (ML) offers significant advantages and considerable potential in studying magnetic materials. Unlike traditional methods, ML does not require predefined theoretical models and can automatically identify nonlinear relationships among complex mechanisms such as magnetocrystalline anisotropy and exchange coupling. This overcomes simplifications inherent in classical models. ML enables multi-scale information integration, combining atomic composition, morphological features, and synthesis conditions. In addition, ML offers high computational efficiency, allowing rapid prediction of key parameters [10,11]. Furthermore, SHAP analysis enables quantitative interpretation of critical factors, providing physical insights for magnetic material design.

In the past three years, machine learning has achieved remarkable success in predicting the performance of magnetic materials. Wang et al. designed cost-effective (Pr,Nd)LaCeFeB permanent magnets [12]. Their design reduced costs by 31.3 % while retaining 86.4 % performance. Yang et al. predicted M_s in Fe-based amorphous alloys with coefficient of determination (R^2) > 0.85 [13]. Guided by this model, they successfully designed novel alloys with record-high M_s , significantly surpassing that of conventional silicon steel. Xu et al. achieved high-precision prediction of Curie temperature for Sm–Co alloys with accuracy exceeding 90 % [14]. This offers a new paradigm for high-temperature permanent magnet design.

However, current ML methods face challenges in magnetic nanoparticles. First, simplified datasets based solely on chemical composition poorly adapt to experimental factors. These include particle size distribution and interfacial defects [15]. Secondly, multi-scale and multi-physics coupling effects are often neglected. This limits predictions of high-temperature magnetism and stress-induced performance [16]. Thirdly, feature engineering has not fully incorporated magnetic exchange mechanisms and electronic structure. This limits predictions for novel materials such as Heusler alloys and low-dimensional magnets [17]. These challenges motivate physics-guided feature engineering, data quality control, and model interpretability.

1.4. Research objectives and contributions

To address these challenges, this study curated a database of 1,758 magnetic nanoparticle samples covering Fe_3O_4 , Fe_2O_3 , and CoFe_2O_4 systems. We systematically compared multiple machine learning algorithms, including Gradient Boosting, XGBoost, and ExtraTrees. Input features included geometric parameters, chemical composition, crystal structure, and synthesis conditions. Gradient Boosting achieved R^2 values of 0.78 and 0.85 for H_c and M_r , while ExtraTrees achieved R^2 of 0.93 for M_s . This framework enables simultaneous multi-parameter prediction, overcoming limitations of previous single-property studies.

Furthermore, SHAP analysis quantitatively identified the dominant mechanisms controlling each parameter. The exchange bias field primarily controls H_c , while the interplay among M_s , H_c , and elemental ratios (Fe, Co, Ni, O) determines M_r . Core magnetic moment and geometric anisotropy govern M_s . These findings establish interpretable links between microscopic features and macroscopic responses, providing guidance for the rational design of magnetic nanoparticles.

2. Methods

2.1. Data source and composition

Based on the performance requirements and theoretical challenges mentioned above, this study selects the three key parameters of the hysteresis loop as target variables for predictive modeling. The hysteresis loop is a fundamental curve that characterizes the magnetic behavior of magnetic nanoparticles. Its key parameters primarily include H_c , M_r , and M_s , which determine the material's performance in different application scenarios [1].

To enable accurate prediction of the above magnetic parameters, this study selected a comprehensive database covering a variety of representative magnetic material types. The database comprises 1,758 samples from two complementary public sources: Kapranova et al. [18] (1,298 samples) and Pavel Kim et al. [19] (460 samples). The Kapranova dataset focuses on exchange bias field and hysteresis parameters of core–shell nanoparticles. The Kim dataset emphasizes M_s for MRI and hyperthermia applications. These datasets compile experimental results recently reported in high-impact international journals and contain variables highly relevant to the objectives of this study.

It should be noted that, although this study utilizes part of the dataset from Kapranova et al., there are fundamental differences in both research objectives and methodology. The work by Kapranova et al. primarily focused on predicting the exchange bias field, a phenomenon that occurs only in ferromagnetic/antiferromagnetic core–shell heterostructures. This represents a specific interfacial coupling magnetic effect, with relatively limited research scope and application scenarios. In contrast, the core objective of the present study is to establish a predictive framework for the three fundamental hysteresis parameters, H_c , M_r , and M_s . These parameters are fundamental physical quantities describing the hysteresis behavior of all types of magnetic nanoparticles, including single-phase nanoparticles, core–shell structures, and multi-layer composites. Therefore, the model developed in this study possesses broader generality and potential applicability.

The dataset used in this study was constructed following explicit selection criteria and quality control procedures to ensure data reliability and representativeness. First, priority was given to experimental data from clearly documented, peer-reviewed literature and public databases to ensure high sample quality, reliability, and scientific rigor of the measurements. Secondly, all data were obtained under a standardized measurement protocol with uniform units, minimizing systematic errors caused by methodological differences. Thirdly, each record contains complete key descriptors, including particle geometry, chemical composition, crystal structure, and synthesis parameters, to ensure that the model captures the main factors influencing magnetic properties. Fourth, representative magnetic material systems were selected to enhance the model's generalization capability across different material spaces.

For quality control, duplicate entries were identified and removed to avoid introducing bias during model training. The plausibility of physical parameters was verified; for instance, magnetization values were required to be positive and fall within reasonable physical ranges. The internal consistency of the dataset was also checked to ensure logical coherence among features. These quality control measures provide a reliable data foundation for subsequent machine learning modeling.

This study utilizes established authoritative databases rather than constructing a fully self-built dataset, for the following reasons: First,

existing databases compile a large volume of high-quality experimental data, covering a wide range of material types and diverse synthesis and measurement conditions, which helps improve the generalization and stability of the model. Secondly, leveraging existing data significantly reduces experimental cost and time, while accelerating model development and validation. In addition, building models based on publicly available data facilitates reproducibility and comparison in subsequent research. Compared to manually collected datasets from literature, avoids issues such as data loss and inconsistency in formatting and standards. The database used in this study offers advantages in terms of data quality, structural consistency, and completeness of experimental conditions, which further enhances data processing efficiency and model applicability. This approach aligns with the trend of open sharing and sustainable development in the field of materials informatics.

The integrated database includes a total of 1,758 samples and covers typical magnetic materials such as Fe_3O_4 , Fe_2O_3 , and CoFe_2O_4 . The H_c and M_r datasets are primarily focused on modeling hysteresis behavior. Their design accounts for the combined influence of structural configuration, particle size, temperature, and interfacial coupling on magnetic properties. The associated features cover a wide range of aspects, including crystal structure, material composition, magnetic response, temperature-related parameters, external magnetic field conditions, geometric morphology, and key electronic structure characteristics.

In contrast, the M_s dataset emphasizes the field-induced magnetic response of nanoparticles. Core features in this subset include the chemical composition of the core and shell, core magnetic moment, particle dimensions, surface modification characteristics, and external excitation conditions.

2.2. Dataset processing and feature engineering

2.2.1. Data cleaning and feature construction

To ensure effective model training and prediction performance, we developed a systematic data processing workflow for each of the three magnetic parameter prediction tasks. This workflow includes data cleaning, feature construction, missing value imputation, and feature selection. Table 1 summarizes the key configuration parameters for the three prediction tasks, including initial feature counts, final feature counts, sample sizes, and feature selection strategies. This table allows readers to quickly compare the data characteristics and feature engineering approaches across different tasks.

To ensure the effectiveness of model training and prediction performance, a systematic data processing workflow was developed for each of the three magnetic parameter prediction tasks, involving data cleaning, feature construction, missing value imputation, and feature selection. For the M_r prediction task, irrelevant fields such as “link” and “id” were removed. The target variable was strictly filtered by excluding missing, non-positive, and non-finite values, and Extreme outliers were removed using the 5th–95th percentile range. This percentile-based method is more robust than standard deviation filtering for magnetic parameters. Extreme values often result from particle aggregation or

measurement artifacts rather than intrinsic magnetic properties.. Chemical formulas were parsed using regular expressions to extract atomic ratios of key elements including Fe, Co, Ni, and O. A volume feature was derived from particle geometry parameters (x, y, z), and temperature-related variables such as the blocking temperature (T_b) and Néel temperature were selected and imputed. Categorical variables such as “core” and “shell” were encoded using frequency encoding. This method assigns numerical values based on material occurrence frequency. It implicitly reflects the experimental maturity and characterization reliability of different compositions. Rare categories were merged into a general “other” class. Finally, correlation and collinearity analyses were conducted to reduce the number of features to ten, ensuring the independence and representativeness of the input variables for high-quality modeling of M_r .

For the M_s prediction task, the raw dataset was loaded and cleaned by removing irrelevant columns. The target variable underwent strict filtering, including the exclusion of null, non-positive, and non-finite values. Extreme outliers were removed by retaining only samples within the 5th to 95th percentile range. Subsequently, feature engineering was conducted. Guided by the Magpie feature construction framework proposed by Ward et al. [20] and the element-based descriptor design strategy of Oliynyk et al. [21], a feature engineering process was implemented. The core and shell compositions were deeply parsed using the ChemicalFormulaEncoder class to extract macroscopic chemical features such as atomic ratios, metallic diversity, total atom count, various physicochemical properties, and mass-to-magnetism ratios. In addition, numerical features such as particle concentration, size, magnetic moment, H_c , and M_r were incorporated. Missing values in these features were imputed using median values. All input features are independently measured physical quantities. No features derived from the target variable M_s were used. Although M_r and H_c come from the same hysteresis loop measurement as M_s , they represent physically independent parameters. which is determined by atomic magnetic moments. which is governed by domain structure. which is determined by magnetic anisotropy. These parameters are controlled by distinct physical mechanisms. No mathematical derivation relationships exist among them. Data leakage was strictly prevented through three measures. First, training and test sets were completely separated. Second, all feature engineering and standardization parameters were computed solely from the training set. Third, the test set was used only for final evaluation, without involvement in model selection or hyperparameter tuning. A multi-stage intelligent feature selection strategy was applied. Features with near-zero variance were removed, followed by elimination of highly collinear features and those with weak linear correlation to the target variable, based on Pearson correlation coefficients. The Magpie-inspired feature engineering approach is physically motivated. M_s is fundamentally determined by atomic magnetic moments from 3d and 4f electron configurations. Feature standardization ensures that parameters contribute to model training based on their predictive importance rather than their numerical scale. Finally, All processed features were then standardized, and the SelectKBest method was used to select the top K features with the highest predictive power, forming the final feature matrix for machine learning model training.

For the H_c prediction task, samples with missing target values were first removed. Subsequently, missing values in other features were imputed according to their specific characteristics. Structural parameters were filled using the mode. Numerical features related to temperature, magnetism, and measurement conditions were imputed with the median. Geometric parameters were prioritized for imputation using K-nearest neighbors (KNN, $k = 5$). This local imputation strategy preserves the physical correlation between particle morphology and synthesis conditions. It maintains the relationship between shape anisotropy and H_c predicted by nucleation theory. When fewer than 3 geometric features were available for neighbor identification, median imputation was used as a fallback. Parameters related to exchange bias were directly filled with zeros to ensure data completeness.

Table 1

Feature configuration and sample size for H_c , M_r , and M_s prediction tasks.

Item	H_c Prediction Task	M_r Prediction Task	M_s Prediction Task
Initial Feature Count	26	26	13
Final Feature Count	11	10	10
Total Sample Size	1298	1298	460
Feature Selection Method	Correlation analysis + weak feature removal	Correlation + collinearity analysis	Variance filtering + correlation analysis + SelectKBest

Feature engineering was conducted based on the given variables, generating derived features such as area and volume. A correlation analysis was subsequently performed to eliminate weakly correlated features, such as shape descriptors and vertical displacement. Finally, 11 representative features were retained for model training.

2.2.2. Statistical distribution analysis

To comprehensively evaluate the effectiveness of data preprocessing and feature engineering, this study conducted systematic distribution characteristic analysis of three key magnetic parameters. Fig. 1 comprehensively displays the statistical distribution characteristics of M_r , H_c , and M_s .

Given the extremely wide range of raw H_c values (0.1–59,144 Oe), logarithmic transformation was applied. This transformation has physical justification. According to the Néel-Brown model, H_c is determined by the energy barrier KV (anisotropy energy $\times$ volume). Both particle

volume and anisotropy constant follow lognormal distributions in synthesized nanoparticles. Therefore, H_c inherently exhibits lognormal behavior. Fig. 1(a) shows the histogram with kernel density estimation (KDE), revealing that the log-transformed data are predominantly distributed within the range of 6–7, exhibiting an approximately normal distribution. Fig. 1b displays the box plot, indicating a median of 6.77 and mean of 6.51, with minimal outliers, demonstrating improved symmetry compared to the original scale. Fig. 1(c) presents the violin plot, which clearly illustrates the concentration of data around the central values, confirming that the transformation effectively resolved the extreme skewness and enhanced modeling stability.

In contrast to H_c , M_r exhibits a pronounced right-skewed distribution, as shown in Fig. 1(d)–(f). The majority of M_r values are concentrated in the low-value range, with the histogram in Fig. 1(d) revealing a sharp decrease in frequency as values increase. This asymmetry is quantitatively confirmed by the box plot in Fig. 1(e), where the median

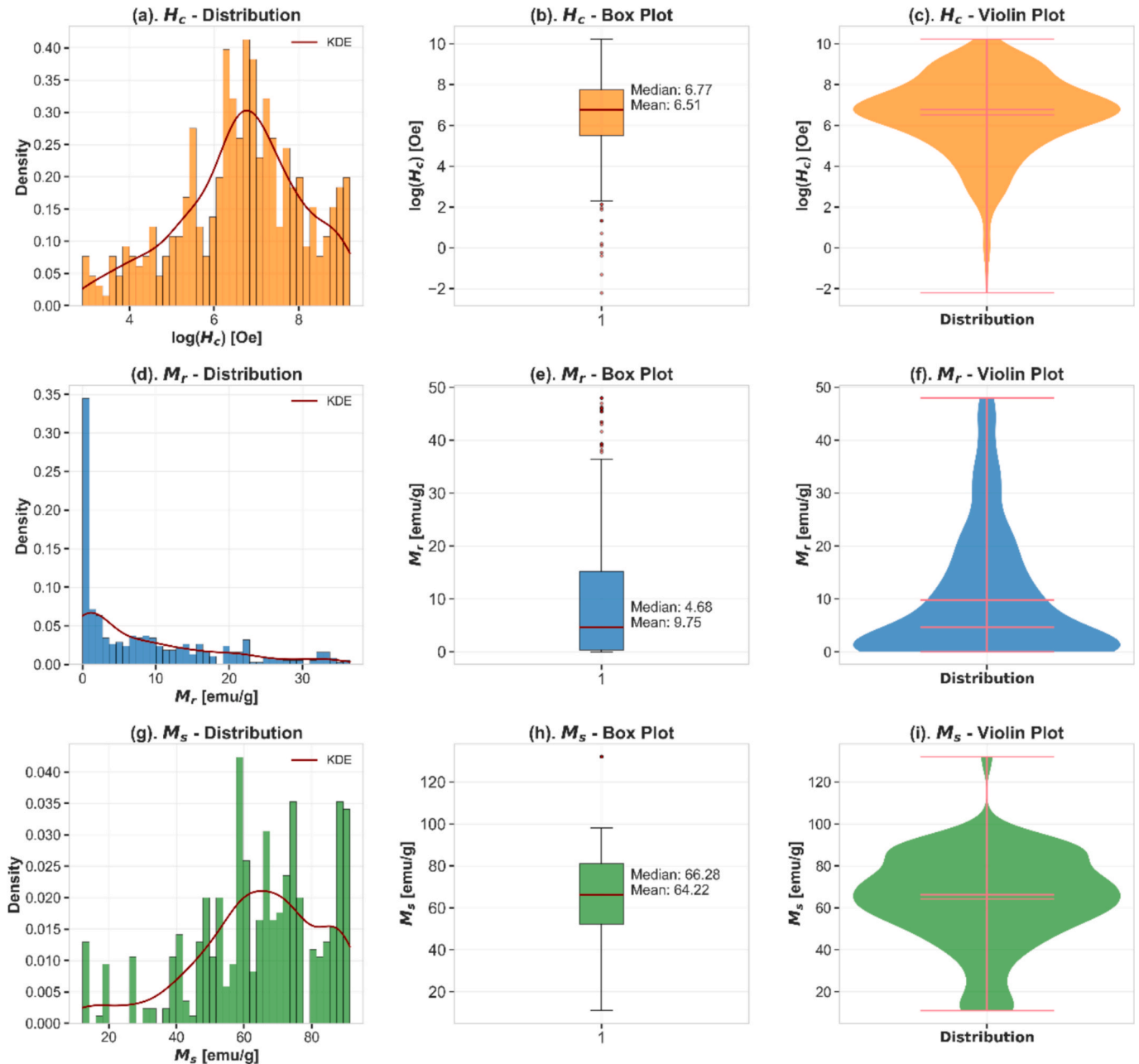


Fig. 1. Statistical distributions of three magnetic parameters: (a)–(c) KDE histogram, box plot, and violin plot for log-transformed H_c ; (d)–(f) KDE histogram, box plot, and violin plot for M_r ; (g)–(i) KDE histogram, box plot, and violin plot for M_s .

(4.68 emu/g) is substantially lower than the mean (9.75 emu/g), indicating the presence of high-value outliers. The violin plot in Fig. 1(f) visually captures this characteristic pattern of dense low values and sparse high values.

Among the three magnetic parameters, M_s displays the most favorable distribution characteristics, as presented in Fig. 1(g)–(i). Unlike the skewed patterns observed for M_r and the transformed H_c , M_s values follow a more balanced distribution centered around 60–80 emu/g (Fig. 1(g)). The close agreement between median (68.28 emu/g) and mean (64.22 emu/g) in the box plot (Fig. 1(h)), combined with the symmetric spindle shape in the violin plot (Fig. 1(i)), indicates strong central tendency and near-Gaussian distribution. This high-quality data structure provides an optimal foundation for subsequent modeling efforts.

Further analysis reveals that all three magnetic parameters exhibit reasonable distribution characteristics after systematic preprocessing. M_r displays a typical right-skewed distribution, with dense low values and sparse high values. H_c significantly improves skewness after logarithmic transformation, approaching symmetry and concentration with greatly enhanced modeling stability; M_s maintains overall balance with slight right skewness, presenting a structure that is wide in the center and narrow at both ends. Comparison of histograms, box plots, and violin plots for the three parameters, the significant differences in their concentration trends, dispersion degrees, and distribution patterns clearly observed.

Analysis of the plots for these three preprocessed parameters provides insights into the distribution characteristics of H_c , M_r , and M_s , including central tendency, dispersion, and distribution shape. These findings offer valuable reference for further investigations into the properties and performance of magnetic materials, and lay a solid foundation for subsequent modeling and analysis based on the data.

2.2.3. Correlation analysis

Following data cleaning, feature construction, and preliminary selection, a correlation analysis was performed on the final feature sets of the three magnetic parameters to validate the effectiveness of feature engineering and gain deeper insights into the internal relationships among variables. Pearson correlation heatmaps were generated to evaluate the linear relationships among model input variables, identify potential multicollinearity risks, and statistically reveal the physical coupling mechanisms between magnetic parameters. The heatmaps use color intensity to indicate correlation strength, with red representing positive correlation and blue indicating negative correlation, ranging from -1 to $+1$. This analysis serves as a key step in quality control of

feature engineering and provides critical support for subsequent model training and interpretation of physical mechanisms [22,23].

Fig. 2 presents the Pearson correlation heatmaps of the predictive features for the three magnetic parameters. Fig. 2(a) illustrates the correlation distribution for H_c prediction, where most coefficients lie within ± 0.6 , indicating that the feature selection strategy effectively mitigates severe multicollinearity. Fig. 2(b) shows the correlation patterns among features for M_r prediction, with composition-related features exhibiting certain trends consistent with the intrinsic stoichiometric constraints, while clear linear relationships are observed between key features and the target variable. Fig. 2(c) displays the correlation distribution for M_s prediction, in which the feature correlations are relatively balanced, with no extreme high correlations, providing a stable data structure for modeling. Overall, the feature sets for all three magnetic parameters are well-structured, retaining critical information while minimizing redundancy. This confirms the effectiveness of prior feature selection and provides a reliable foundation for subsequent model training and interpretation of the underlying physical mechanisms [24,25].

To ensure reproducibility, this study provides detailed software version information. All data analysis and modeling were performed using Python 3.13.0. Machine learning models were implemented using Scikit-learn 1.7.1. SHAP 0.49.1 was used for model interpretability analysis. NumPy 2.3.4 and Pandas 2.3.3 were employed for data processing. Matplotlib 3.10.7 was used for data visualization. The dataset was split into training and testing sets using an 80:20 ratio. Model performance was evaluated using 5-fold cross-validation. Hyperparameter optimization was performed through grid search combined with 5-fold cross-validation. Detailed model configurations and performance results are presented in Section 3.1.

3. Results and discussion

3.1. Model prediction performance and algorithm comparison

To comprehensively evaluate different modeling strategies for predicting magnetic parameters, this study systematically selected six representative regression models. These include two conventional linear models—Linear Regression and ElasticNet—as well as four widely used tree-based nonlinear ensemble methods: Random Forest, ExtraTrees, Gradient Boosting, and XGBoost.

Linear models, due to their simple structure, high computational efficiency, and inherent interpretability, are suitable for problems where the relationship between features and target variables is approximately

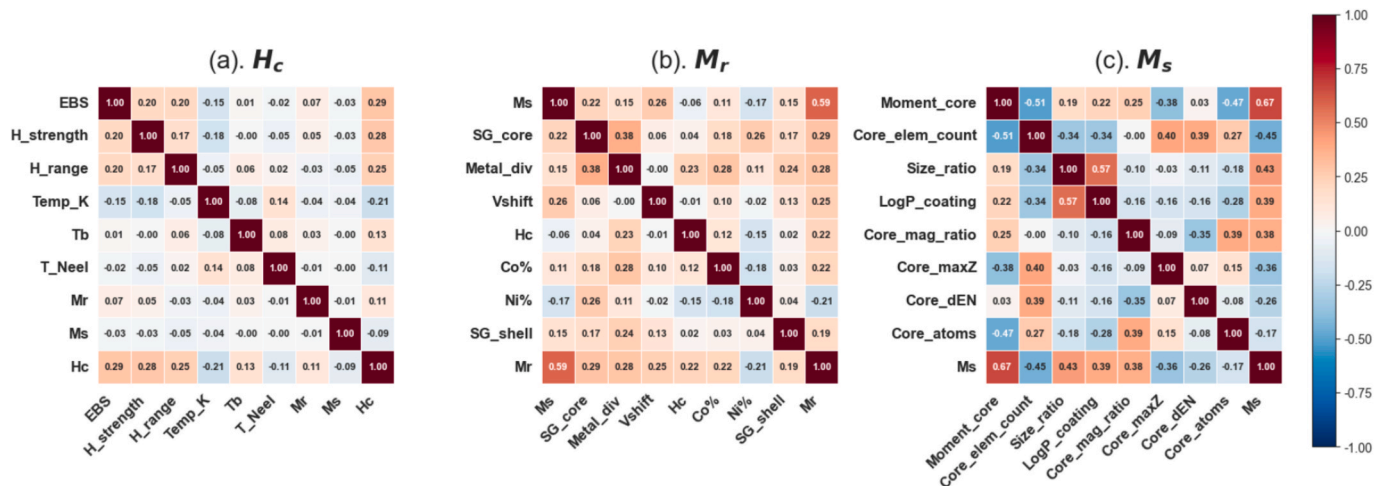


Fig. 2. Correlation heatmaps of predictive features for three magnetic parameters. (a) Correlation matrix for H_c prediction; (b) Correlation matrix for M_r prediction; (c) Correlation matrix for M_s prediction.

linear. However, they often fail to adequately capture the underlying patterns when the data involve complex nonlinear associations or significant feature interactions. In contrast, tree-based ensemble methods possess strong nonlinear fitting capabilities and can automatically model nonlinear relationships and interactions among high-dimensional features. They also exhibit robustness to outliers and variations in feature scales, making them widely applied in property prediction tasks in materials science.

This study also explored deep learning models, such as neural networks; however, due to the limited sample size and the high dimensionality of features, these models were prone to overfitting during training, resulting in suboptimal prediction accuracy and stability. The results indicated that neural networks were less effective at capturing the complex nonlinear relationships and multi-scale feature interactions inherent in magnetic nanoparticles. In contrast, tree-based ensemble models demonstrated greater robustness in handling nonlinear relationships and high-dimensional feature interactions, thereby revealing underlying physical mechanisms and avoiding the limitations of “black-box” approaches. Furthermore, tree-based ensemble methods offer high computational efficiency and fast training speed, making them well-suited for rapid iteration and parameter optimization in materials design. Based on these comparative results, we ultimately selected tree-based ensemble models as the primary modeling strategy, which exhibited significant advantages in predicting the magnetic properties of nanoparticles.

Therefore, the model performance in this study was systematically evaluated using the coefficient of determination R^2 and root mean squared error RMSE on both the training and testing sets. The results are presented in Table 2.

In the joint modeling of three key magnetic parameters— H_c , M_r , and M_s —tree-based ensemble models such as Gradient Boosting, XGBoost, and ExtraTrees significantly outperformed traditional linear regression models in both accuracy and stability (as shown in Table 2). These models demonstrated excellent nonlinear modeling capability and generalization performance. Gradient Boosting achieved the highest test coefficient of determination in predicting H_c , M_r , with R^2 values of 0.78 and 0.85, respectively. For the M_s prediction task, ExtraTrees performed best, achieving a test R^2 as high as 0.93. This confirms the broad applicability of ensemble methods based on random feature subsets in magnetic parameter prediction. The mean absolute error (MAE) values further validated model accuracy: 0.58 for H_c , 2.66 for M_r (both using Gradient Boosting), and 2.93 for M_s (using Extra Trees).

By comparison, ElasticNet showed significantly poorer predictive performance across all three tasks, with R^2 values mostly below 0.60. Specifically, for H_c , M_r prediction, its R^2 values were only 0.34 and 0.40, respectively, indicating that the linear structure failed to adequately capture the complex nonlinear physical mechanisms underlying the magnetic behavior of the materials. This further confirms the high sensitivity of magnetic parameter prediction to the complexity of modeling algorithms and the expressive power of features.

3.2. Optimal model hyperparameter configurations

To ensure reproducibility, this study provides detailed optimal

Table 2

Performance comparison of different machine learning models in predicting three magnetic parameters (H_c , M_r , and M_s).

model/metric	H_c R^2	H_c RMSE	M_r R^2	M_r RMSE	M_s R^2	M_s RMSE
GradientBoosting	0.78	0.86	0.85	4.30	0.90	7.72
LightGBM	0.73	0.95	0.84	4.47	0.83	9.57
XGBoost	0.78	0.86	0.83	4.52	0.92	6.71
ExtraTrees	0.73	0.96	0.80	4.91	0.93	6.01
RandomForest	0.73	0.96	0.78	5.15	0.92	6.56
ElasticNet	0.10	1.84	0.40	8.53	0.65	13.71

hyperparameter configurations for each prediction task. These hyperparameters were optimized through grid search combined with 5-fold cross-validation. For the H_c prediction task, the Gradient Boosting model employed an optimal configuration with 120 weak learners, a maximum tree depth of 7, a learning rate of 0.08, a sample sampling ratio of 0.7, a minimum of 5 samples required for node splitting, a minimum of 3 samples required for leaf nodes, and a random seed of 6000. For the M_r prediction task, the Gradient Boosting model used 100 weak learners, a maximum tree depth of 5, a learning rate of 0.1, a sample sampling ratio of 0.8, a minimum of 4 samples for node splitting, a minimum of 2 samples for leaf nodes, and a random seed of 42. For the M_s prediction task, the Extra Trees model utilized 200 weak learners with no limit on tree depth, a minimum of 2 samples for node splitting, a minimum of 1 sample for leaf nodes, and a random seed of 42. These configurations achieved optimal performance for their respective tasks, and the fixed random seeds ensure full reproducibility of results.

Based on the comprehensive performance evaluation, Gradient Boosting was identified as the most advantageous modeling algorithm in this study. Gradient Boosting is a powerful ensemble learning method that iteratively trains a series of weak learners, typically decision trees, and combines them to form a strong learner. It optimizes the model stepwise by minimizing the gradient of the loss function, with each iteration learning from the residual errors of the previous model. This allows it to effectively capture the complex nonlinear relationships and multidimensional feature interactions in the magnetic behavior of nanoparticles [26,27]. Its advantages lie in accurately modeling magnetic behaviors influenced by the complex coupling of multiple physical mechanisms. It can effectively handle diverse features including geometric parameters, chemical composition, crystal structure, and synthesis conditions. It exhibits certain robustness to data noise and, through tools like SHAP, achieves high model interpretability by quantifying the contribution of each feature to the predictions, thereby revealing underlying physical mechanisms.

To further verify the stability and generalization ability of the constructed models, this study conducted a systematic evaluation of the prediction tasks for three magnetic parameters using five-fold cross-validation.

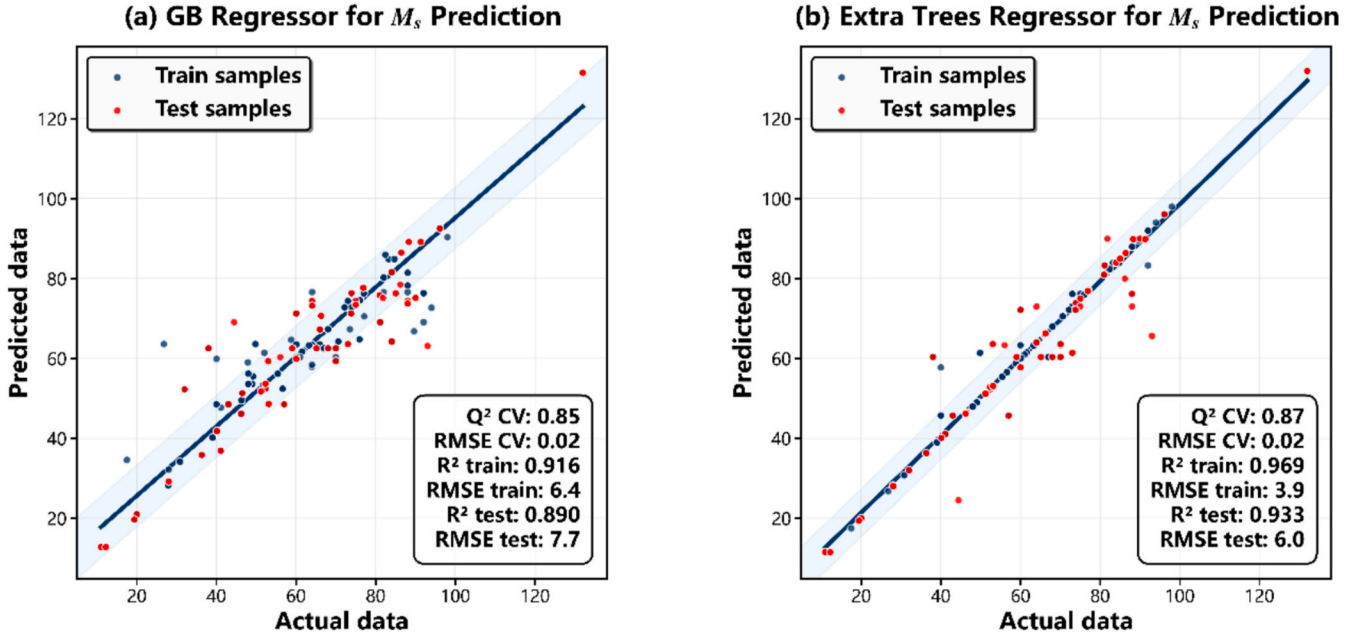
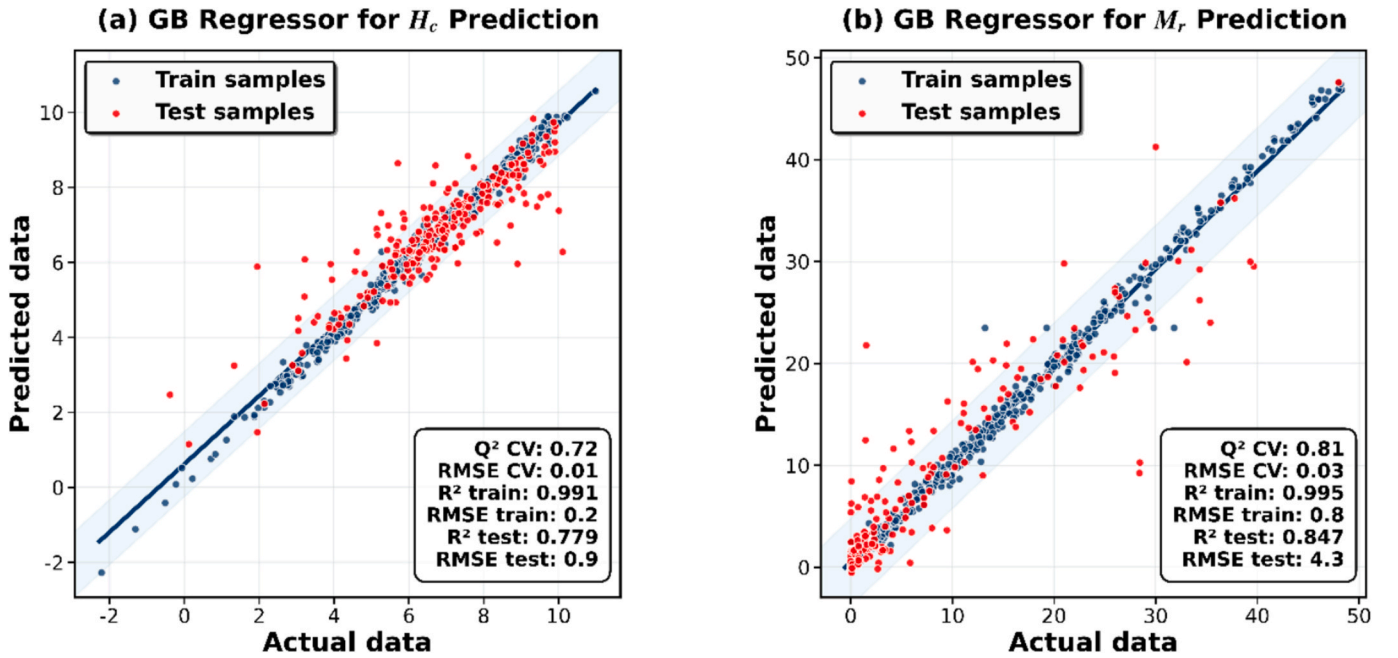
Fig. 3 presents the five-fold cross-validation results of predicting the M_s using Gradient Boosting and Extra Trees regressors. The horizontal axis represents the actual values, while the vertical axis represents the predicted values. Blue dots indicate training samples, and red dots indicate testing samples. Each subplot includes the fitted regression line and confidence interval. Fig. 3(a) shows the prediction results of the Gradient Boosting regressor, and Fig. 3(b) shows those of the Extra Trees regressor.

The cross-validation results indicate that both models achieve a mean R^2 above 0.80, demonstrating good predictive consistency. Notably, the Extra Trees regressor exhibits superior performance in predicting M_s , with a higher R^2 value indicating its stronger ability to capture complex relationships in the data and better generalization capability.

Fig. 4 presents the prediction results of H_c and M_r using the Gradient Boosting regressor. Fig. 4(a) shows the prediction of H_c . Although the mean R^2 for H_c is only 0.72, slightly lower than that for M_r and M_s , the predicted points generally align along the diagonal, indicating that Gradient Boosting can capture the variation trend of H_c to a certain extent. The differences between training and testing results are limited, suggesting that the model possesses reasonable fitting and generalization abilities, though prediction accuracy can still be improved.

Fig. 4(b) shows the prediction of M_r . The predicted points closely follow the diagonal, with both training and testing samples distributed concentrically. The cross-validation R^2 exceeds 0.80, notably higher than that for H_c , indicating that the model demonstrates superior stability and generalization ability in predicting M_r , with more robust prediction performance.

The relatively lower prediction performance for H_c is mainly

Fig. 3. Five-fold cross-validation results for the prediction of M_s .Fig. 4. Five-fold cross-validation results for the prediction of H_c and M_r .

attributed to its complex physical mechanisms and broad numerical distribution. Although logarithmic transformation was applied to partially mitigate the influence of extreme values, the original data still showed a wide range. Furthermore, H_c is affected by the coupling of multiple nonlinear factors, such as exchange bias field, particle geometry, and magnetic anisotropy. These factors make the modeling of H_c significantly more difficult than that of remanent and M_s . This complexity limits further improvements in model accuracy to some extent, but the overall cross-validation results still confirm the effectiveness of the models in the H_c prediction task.

In summary, this study systematically compared several mainstream modeling approaches regarding their applicability and performance in predicting magnetic parameters. The results confirm the clear

advantages of nonlinear ensemble tree models in capturing complex magnetic behaviors. Gradient Boosting and ExtraTrees models achieved leading predictive performance in different tasks, demonstrating excellent accuracy and generalization ability. These models provide robust tools and theoretical support for the performance optimization of high-performance magnetic nanomaterials.

3.3. Feature importance analysis and physical mechanism interpretation

To thoroughly reveal the intrinsic influencing factors and underlying physical mechanisms of key magnetic parameters in magnetic nanoparticles, this study conducted a comprehensive variable importance and contribution analysis based on the SHAP method [28]. By

quantitatively assessing the impact of each input feature on the predictions of H_c , M_r , and M_s , the SHAP analysis not only revealed the core physical driving factors [29] but also validated the coupling relationships among parameters. This data-driven interpretability analysis provides scientific insights into the microscopic mechanisms underlying complex magnetic behaviors and lays a solid foundation for the rational design and performance optimization of magnetic nanoparticles [30]. The mean absolute SHAP value is defined as:

$$|\overline{\phi_i}| = 1/N \sum_{j=1}^N |\phi_i^{(j)}| \quad (1)$$

where N is the total number of samples, and $\phi_i^{(j)}$ represents the SHAP value of feature i for the j sample. This metric reflects the average magnitude of the feature's impact on the prediction, with larger values indicating higher importance of the feature in the model. The sign of the SHAP value indicates whether the feature contributes to increasing or decreasing the predicted target. By utilizing SHAP analysis, sample-level interpretability of model predictions can be achieved, revealing the contribution weight of different features for specific predictions, thereby providing scientific insights for the rational design and performance optimization of magnetic nanoparticles.

3.3.1. SHAP analysis for H_c

Fig. 5 presents the SHAP analysis results for H_c prediction based on the Gradient Boosting model, illustrating the contribution of each feature to H_c predictions. The regulation mechanism of H_c goes beyond simple intrinsic magnetism. SHAP analysis identified the exchange bias field as a key controlling factor for H_c . As shown in Fig. 5, this feature exhibits the highest mean absolute SHAP value of 0.56 in the Gradient Boosting model, indicating its dominant contribution to the model's predictions. Further statistics reveal that approximately 54 % of all samples have positive SHAP values for this feature, suggesting a positive correlation between the exchange bias field and H_c . This finding is

highly consistent with the theoretical studies by Iglesias et al. [31] and De et al. [32], on the interfacial exchange coupling mechanism in core-shell structured nanoparticles. These studies show that interface coupling can enhance the energy barrier for domain wall reversal, effectively increasing the H_c and magnetic stability of the material, thus explaining the dominant role of the exchange bias field in H_c variation observed in the SHAP analysis.

The underlying physical mechanisms can be further understood through established magnetic theories. Surface effects in nanoparticles originate from broken symmetry and reduced coordination at interfaces. These factors modify the local magnetic anisotropy energy. The Néel surface anisotropy model shows that interfacial spins contribute to the effective magnetic anisotropy. This contribution scales with the surface-to-volume ratio [33,34]. In core-shell nanoparticles, exchange coupling occurs at the ferromagnetic-antiferromagnetic interface. This coupling creates unidirectional anisotropy, manifesting as exchange bias. The coupling strength depends on interfacial spin configuration, magnetic anisotropy of both phases, and interfacial structural quality [35,36]. These mechanisms explain the SHAP-identified dominant role of exchange bias in controlling H_c .

Besides the exchange bias field, the model also identified the maximum applied magnetic field range as another key variable, which has a positive influence on H_c prediction. The SHAP results indicate a robust positive correlation between this feature and H_c , suggesting that higher applied magnetic fields help drive magnetic nanoparticles through a complete hysteresis loop process, triggering more domain reversal behaviors and interfacial response mechanisms. This conclusion aligns with the observations of Skumryev et al. [37] in ferromagnetic/antiferromagnetic (FM/AFM) nanoparticle systems, where enhanced external fields were found to improve magnetic stability and H_c , thereby enabling the system to “overcome” the superparamagnetic limit.

3.3.2. SHAP analysis for M_r

Fig. 6 shows the SHAP analysis results for M_r prediction using the

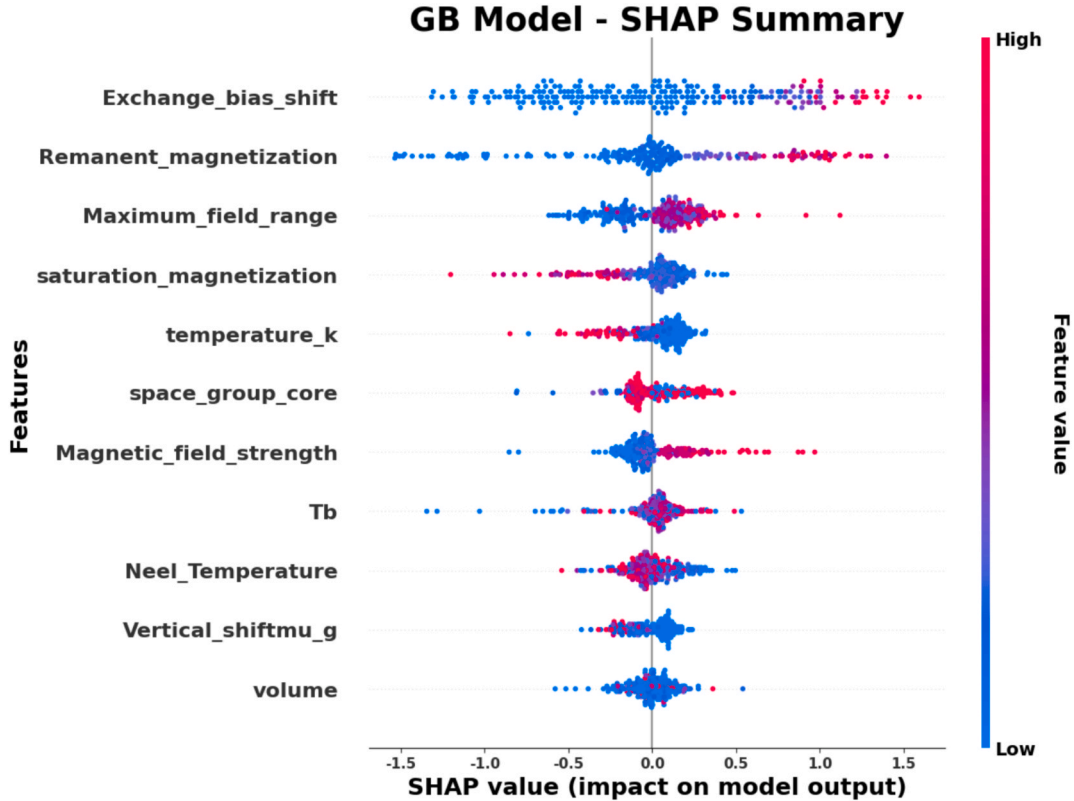


Fig. 5. SHAP summary plot for H_c .

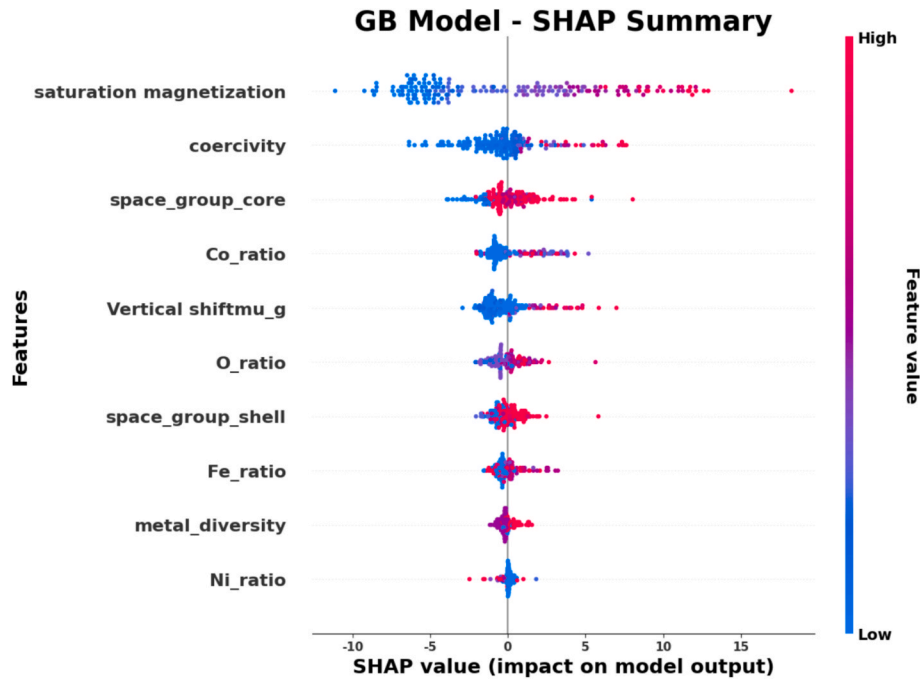


Fig. 6. SHAP summary plot for M_r .

Gradient Boosting model, indicating the varying importance and impact direction of key features in predicting M_r . For M_r , M_s exhibits the most significant and dominant positive effect in the M_r prediction model. As shown in Fig. 6, this feature has a mean absolute SHAP value of 5.40 in the Gradient Boosting model, exceeding those of other features. The SHAP plot distribution shows that red dots, representing samples with high feature values, are prominently clustered in the positive region. This observation intuitively indicates that the higher the M_s of the material, the greater the M_r . This finding is highly consistent with the classical Stoner-Wohlfarth model, which states that for single-domain particles, there is a direct proportional relationship between M_r and M_s (with a ratio close to 0.5) [6], reflecting M_r as the residual effect of M_s at zero field.

H_c also shows a significant positive contribution in the model, with SHAP analysis further quantifying its impact. With a mean SHAP value of 1.84, H_c represents the material's resistance to magnetization reversal. Although its overall influence is less than that of M_s , most SHAP values are positive in the region corresponding to samples with high H_c . High H_c means that the magnetization direction is less susceptible to thermal fluctuations or demagnetizing fields after the external magnetic field is removed, thereby helping to maintain stable magnetic moments and effectively sustain high M_r . This physical correlation is widely observed experimentally [38], and SHAP analysis further quantifies its contribution.

Beyond the magnetic parameters themselves, chemical composition also has a significant impact on M_r . SHAP analysis shows that the atomic ratios of elements such as Co, Fe, Ni, and O contribute significantly to M_r prediction, with mean SHAP values ranging from 0.22 to 5.40. These elements form the magnetic framework of the nanoparticles, and their content and interactions determine the intrinsic magnetic moment, exchange interactions, and magnetic anisotropy of the material, making them key determinants of M_s and M_r [39,40].

In summary, the formation and retention of M_r result from the combined effects of multiple factors, with M_s and H_c being the primary macroscopic determinants. At the microscopic level, crystal structure and chemical composition fundamentally regulate M_r by influencing the intrinsic magnetism and magnetization reversal mechanisms of the material. These SHAP analysis results provide data-driven and

interpretable physical insights for the design and performance optimization of magnetic nanoparticles.

3.3.3. SHAP analysis for M_s

Fig. 7 presents the SHAP analysis results for M_s prediction using the Extra Trees model, clearly demonstrating the contributions of features to M_s and the model's decision-making mechanism. For M_s , as shown in Fig. 7, higher core magnetic moments (red dots) are strongly associated with larger positive SHAP values, indicating that the magnitude of the core material's magnetic moment is the most critical factor determining the M_s of nanoparticles. Its mean absolute SHAP value reaches 6.21, far exceeding other input variables, demonstrating its central role in model prediction. When the core magnetic moment exceeds -0.33 , all corresponding samples exhibit positive SHAP contributions, indicating that the magnetic moment of core magnetic atoms directly determines the material's saturation response under an applied magnetic field. This conclusion aligns with classical magnetic theory models, which state that the M_s is given by the total magnetic moment of all magnetic atoms divided by the volume V . Both the number of magnetic atoms and their moments contribute to increasing M_s [41], thus providing the physical basis for the core magnetic moment as a key driving factor.

Besides the core magnetic moment, particle structural morphology also shows significant influence. The model identified the maximum-to-minimum size ratio of particles as a nonlinear regulatory feature, with SHAP values ranging from -5.07 to 13.98 , showing positive or negative contributions in different samples. This phenomenon is closely related to nanoscale size effects and shape anisotropy, where non-spherical particles tend to induce local magnetic anisotropy fields, thereby affecting domain formation and magnetization processes [42]; meanwhile, an increased proportion of surface atoms enhances surface disorder and magnetic disturbances [18], making size and shape indirect factors in regulation.

Further SHAP analysis reveals the role of chemical composition in the prediction model. The most abundant core element shows a mean SHAP value of 2.35, reaching up to 4.34 in some samples. Additionally, physicochemical properties such as core magnetic ratio, core electronegativity range, and the number of shell atoms exhibit significant contributions in certain samples. Notably, the electronegativity range,

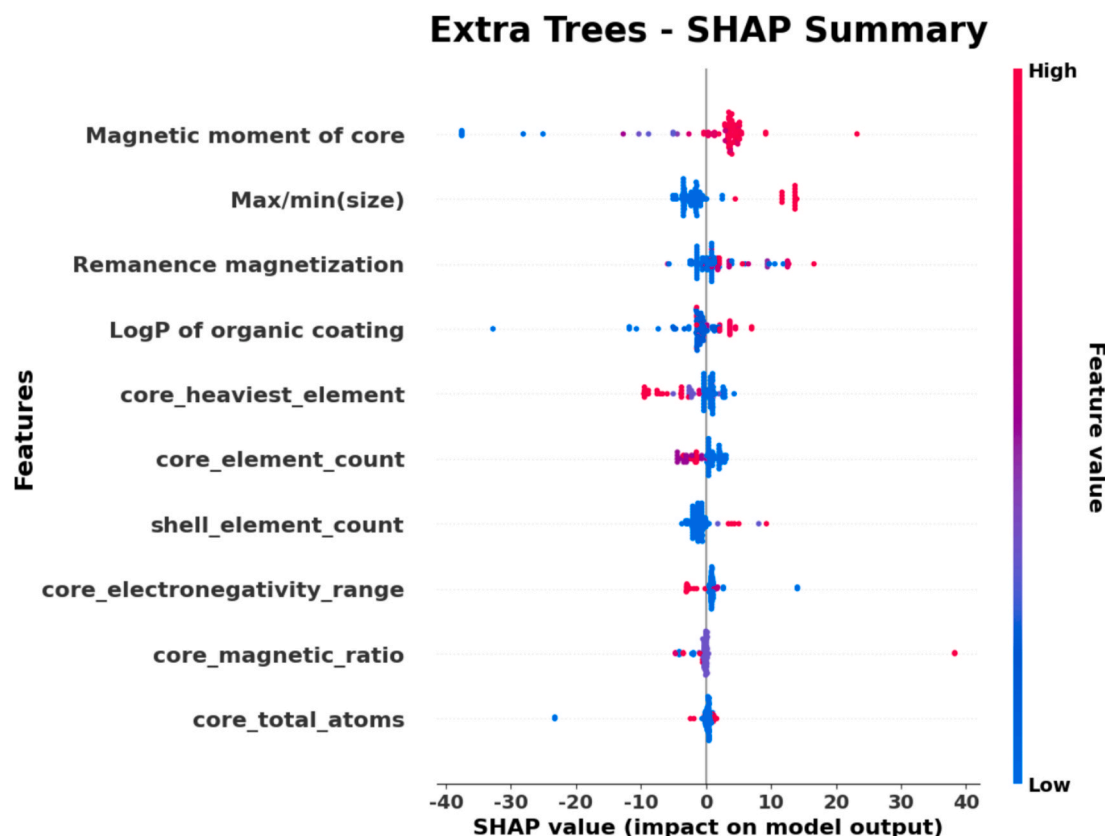


Fig. 7. SHAP summary plot for M_s .

as an indicator of local electronic structure complexity, may influence the cooperative alignment of magnetic moments by modulating inter-orbital transitions and electron coupling strength. This mechanism can be explained by Anderson's superexchange model [43] and the Goodenough-Kanamori rules [44], indicating that microscopic electronic structure parameters can regulate macroscopic M_s by affecting magnetic exchange pathways.

In summary, this study integrates SHAP interpretability analysis with physical model derivations to systematically reveal the multiscale factors affecting M_s , confirming the dominant role of the core magnetic moment as a decisive factor and clarifying the synergistic regulatory mechanisms of structural morphology, chemical composition, and electronic structure on M_s . These SHAP-derived insights align with fundamental magnetic theory. M_s is determined by the net magnetic moment per unit volume. This quantity depends on three factors: the number of magnetic atoms, their individual moments, and the degree of magnetic ordering. Shape anisotropy arises from magnetostatic energy. It influences domain structure and magnetization processes. This effect is particularly important in non-spherical particles because demagnetizing fields vary with direction [45–46]. Electronic structure also plays a key role. Spin-orbit coupling and crystal field interactions modulate local magnetic moments and exchange interactions. These effects provide the microscopic origins for compositional dependencies [47]. This integration of SHAP analysis with physical models serves two purposes: validating machine learning predictions and providing mechanistic guidance for designing and optimizing nanomagnetic materials.

3.3.4. Validation of SHAP results against experimental literature

Comparing SHAP feature importance rankings with experimental literature quantitatively is challenging because traditional experiments often use single- or few-factor designs. Therefore, indirect validation using qualitative literature evidence is necessary.

For H_c , the SHAP-identified dominance of exchange bias aligns with

established understanding of core-shell nanoparticles, where interfacial exchange coupling is a key mechanism for enhancing H_c [31,32,35]. Our model's secondary ranking of the external magnetic field is also consistent with experimental reports that its effect is comparatively weaker [37].

For M_r , its primary dependence on M_s is a direct consequence of the Stoner-Wohlfarth model [6] a relationship extensively observed in standard magnetic nanoparticle systems like Fe_3O_4 and CoFe_2O_4 [36]. The secondary stabilizing role of H_c [33,34], and the fundamental influence of chemical composition on magnetic moment [38–40] are also captured by our SHAP analysis, whereas traditional studies seldom rank these factors systematically.

For M_s , the SHAP ranking underscores the core magnetic moment as the decisive factor, reflecting the basic physical definition of M_s [41]. The complex, non-monotonic influence of particle shape revealed by SHAP is attributed to the competing effects of shape anisotropy and surface spin disorder, both of which are well-documented in experimental studies [42,46,47].

Although SHAP rankings generally align with established literature, limitations exist. Traditional experiments lack complete feature rankings, limiting comparisons to qualitative alignment. Some SHAP-important features (e.g., electronegativity) have clear theoretical foundations [43,44] but lack explicit experimental ranking. Differences in material systems and synthesis conditions may also cause local deviations between model predictions and specific experimental results.

4. Conclusion

4.1. Overview of research findings

This study addresses the scientific challenges in predicting the magnetic parameters of magnetic nanoparticles by proposing a machine learning-based multi-parameter prediction framework. This approach

overcomes key limitations of traditional theoretical models including computational cost, oversimplified physical mechanisms, and limited generalization ability. Using data from typical material systems such as Fe_3O_4 , Fe_2O_3 , and CoFe_2O_4 , the model demonstrates robust performance and adaptability across different materials in predicting three key parameters: H_c , M_s , and M_r .

SHAP feature importance analysis quantitatively reveals the dominant influences of magnetic atom content, exchange bias field, and geometric anisotropy on magnetic performance. Specifically, the exchange bias field is the primary factor influencing H_c , while the content of magnetic atoms (e.g., Fe and Co), atomic ratios of oxygen and nickel, and metal diversity mainly determine M_r . Meanwhile, particle geometric anisotropy influences both M_r and H_c by modulating magnetization reversal mechanisms. For M_s , the core magnetic moment is the most critical intrinsic determinant, influenced jointly by chemical complexity and crystal structure. Compared with previous studies that primarily focus on predicting a single magnetic parameter, such as the exchange bias field, this study achieves, for the first time, the simultaneous prediction of the three fundamental hysteresis parameters (H_c , M_r , and M_s) and reveals their intrinsic coupling relationships, providing a systematic approach for multi-objective, synergistic optimization of magnetic nanoparticles. These results not only confirm existing experimental findings but also extend the understanding of complex magnetic mechanisms.

This study demonstrates multiple advantages of machine learning in magnetic nanomaterials research. It uncovers complex relationships between magnetic parameters and material features, identifying mechanistic factors that traditional theories struggle to quantify, such as magnetocrystalline anisotropy, surface effects, and exchange coupling. Meanwhile, by integrating multiple model frameworks and interpretability analysis, it achieves physical mechanistic tracing of predictions, avoids “black-box” outcomes and enhances the scientific credibility of the models.

In summary, the proposed magnetic parameter prediction model for magnetic nanoparticles overcomes the bottlenecks of traditional methods in describing complex nanoscale mechanisms. It establishes an interpretable bridge from atomic composition and structural features to macroscopic magnetic responses, providing new insights for rational design, performance control, and intelligent optimization of magnetic nanomaterials. This contributes to improving the efficiency and depth of magnetic behavior analysis in complex systems. The employed methods show strong synergy in performance prediction, factor identification, and model interpretation, providing valuable references for parameter screening and performance optimization in broader material systems.

4.2. Directions for future work

Although this study has achieved strong performance in predicting M_r , H_c , and M_s , and SHAP analysis has revealed the dominant contributions of key features, improvements are still needed in model interpretability and in aligning predictions with underlying physical mechanisms. Therefore, future work will adopt multiple strategies to further enhance the physical reliability and interpretability of the predictions. First, local interpretability methods such as LIME (Local Interpretable Model-agnostic Explanations) will be employed to provide more detailed explanations of the predictions for individual samples, facilitating the understanding of the unique behavior patterns of specific material systems. Secondly, targeted experiments will be designed to validate the predicted contributions of key features. The experimental results will then be used to assess and optimize model performance, establishing a closed-loop validation between predictions and experimental data and further improving model accuracy and reliability.

Future research will build upon the current findings and extend in multiple cutting-edge directions. Regarding novel material systems, the predictive framework will be expanded to emerging magnetic nanomaterials. These include two-dimensional materials with unique

electronic structures (e.g., CrI_3 , Fe_3GeTe_2), compositionally complex high-entropy alloy nanoparticles, exchange-coupled composites exhibiting interfacial coupling effects, and rare-earth-based nanoparticles with significant potential in permanent magnet applications. In terms of inverse design capabilities, intelligent optimization algorithms will be developed to automatically determine the optimal nanoparticle composition, structural configuration, and synthesis conditions based on specified target magnetic properties. This approach shifts the design approach from traditional trial-and-error to an efficient, performance-driven workflow.

In terms of methodological innovation, the integration of molecular dynamics simulations, micromagnetic numerical calculations, and machine learning predictions will provide a more powerful theoretical toolkit for understanding and predicting complex magnetic phenomena. Implementation of these research directions is expected to further promote rational design, intelligent manufacturing, and efficient application of magnetic nanomaterials, providing critical support for technological advancements in the field.

4.3. Prospective applications

Leveraging the high-precision prediction capability for magnetic parameters, the framework developed in this study demonstrates strong potential in both biomedical and data storage applications. In the biomedical field, the model can guide the rational design of magnetic hyperthermia nanoparticles by predicting and optimizing the combination of H_c and M_s . This enables precise control of the specific absorption rate (SAR) and provides personalized tumor hyperthermia treatment. In MRI contrast agent development, the model can inversely design nanoparticles with specific M_s values to achieve the desired relaxation enhancement, improving imaging quality and diagnostic accuracy. For targeted drug delivery, balancing M_r and H_c allows for precise navigation and controlled release of nanoparticle-based drug carriers under in vivo magnetic field gradients.

In the field of data storage, this predictive framework can guide the design of high-density magnetic storage media by predicting and tuning nanoparticles with high H_c and moderate M_s , significantly enhancing storage density and thermal stability. Additionally, fine-tuning the M_r/M_s ratio can optimize the read/write speed and signal strength of magnetic storage devices, meeting the demands of next-generation high-performance storage technologies. The model can also facilitate the design of multilayer magnetic structures with varying H_c values, enabling multibit information storage and advanced data encoding functions.

4.4. Research limitations and mitigation strategies

Although this study has achieved significant progress, several inherent limitations remain. First, regarding dataset representativeness, the current database covers major magnetic nanoparticle systems such as Fe_3O_4 , Fe_2O_3 , and CoFe_2O_4 . However, its coverage of emerging materials, including Heusler alloys and two-dimensional magnetic materials (e.g., CrI_3 , Fe_3GeTe_2). Furthermore, the dataset primarily reflects magnetic properties dominated by superexchange interactions in oxides; potentially under-representing mechanisms crucial in other material classes, such as the high magnetocrystalline anisotropy in L1₀-phase alloys (e.g., FePt) or complex interfacial coupling in synthetic antiferromagnets. To mitigate this limitation, future work will focus on gradually expanding the database and constructing material-specific sub-models to extend the applicability of the predictive framework.

Secondly, concerning model generalization, five-fold cross-validation results indicate robust internal generalization within the studied material space ($R^2 = 0.78\text{--}0.93$). However, extrapolating to materials with substantially different compositions (e.g., rare-earth-based nanoparticles) still requires experimental validation. Ongoing and future experiments will be used to assess and improve the model's

extrapolative performance.

Lastly, regarding overfitting control, regularization techniques and early stopping strategies were employed to prevent overfitting. Learning curve analysis demonstrates that training and validation errors converge, confirming the appropriateness of the model complexity. Nevertheless, for sub-datasets with relatively small sample sizes, more cautious model selection and validation strategies are still necessary.

CRedit authorship contribution statement

Yifan Chen: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Yapeng Zhang:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis. **Xinxin Zhang:** Writing – review & editing, Validation, Supervision, Project administration, Formal analysis, Data curation. **Shuangbao Ma:** Writing – review & editing, Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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